Analysing the Wilson–Cowan Neural Field model

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The model

The Wilson–Cowan (WC) model is a classical mathematical model in computational neuroscience describing the interaction between an excitatory neuronal population and an inhibitory neuronal population [1, 2].

Let $E(t)$ denote the activity of the excitatory population and $I(t)$ the activity of the inhibitory population. A common form of the WC equations is:

$$\tau_E \frac{dE}{dt} = -E + \phi(w_{EE}E - w_{EI}I + P)\,, \tag{1}$$

$$\tau_I \frac{dI}{dt} = -I + \phi(w_{IE}E - w_{II}I + Q)\,, \tag{2}$$

where $\tau_E, \tau_I > 0$ are time constants, $w_{EE}, w_{EI}, w_{IE}, w_{II}$ are coupling coefficients, P and Q represent external inputs, and $\phi(\cdot)$ is a nonlinear activation function (e.g. a sigmoid).

Threshold To receive a pass mark for this project, you must complete the following tasks:

- Read the relevant literature on the Wilson–Cowan model and discuss the various terms in the model, including the role of excitation, inhibition, coupling strengths, and external inputs on the dynamics of the system.
- Write code to numerically solve the WC model using a suitable numerical scheme (e.g., a 4th order Runge–Kutta time stepping scheme) and stuy the behaviour of the model for different parameter values (e.g., different coupling strengths, external inputs, and time constants) and critically discuss your observations.
- Extend the Wilson–Cowan model into a *neural field model*, where the excitatory and inhibitory activities depend on both time t and spatial location x. Here, you could consider a two-dimensional grid (of size N) and use vectors

  $$\mathbf{E}(t), \mathbf{I}(t) \in \mathbb{R}^N$$

  to denote the excitatory and inhibitory activity over the grid positions and matrices

  $$W_{EE}, W_{EI}, W_{IE}, W_{II} \in \mathbb{R}^{N\times N}$$

  to describe the connectivity between spatial positions.
- Numerically solve the neural field model for a range of parameter values and initial conditions. Plot and discuss the resulting spatiotemporal behaviour. In your discussion you could for example include: how the behaviour changes as the spatial coupling strength or locality is varied.

Higher Here, you will combine the neural field Wilson–Cowan model with a machine learning task (classification of handwritten digits). The aim is to investigate whether the neural field dynamics can be used as a feature extractor for digit classification by making the connectivity weights trainable and learning the connectivity matrices (or a parameterisation of these matrices) using gradient-based optimisation.

You will use the MNIST dataset of handwritten digits (classes 0–9). Each image can be represented as an input vector

$$u \in \mathbb{R}^N,$$

where $N = 28 \times 28$ is the number of spatial locations and u_i corresponds to the intensity at position x_i. For each digit, the image u is injected as external input into the neural field, and the Wilson–Cowan equations are integrated up to time T. The final excitatory and inhibitory states can then be used as features for classification.

References

[1] Jack D Cowan, Jeremy Neuman, and Wim van Drongelen. "Wilson–Cowan equations for neocortical dynamics". In: *The Journal of Mathematical Neuroscience* 6.1 (2016), p. 1.

[2] Hugh R Wilson and Jack D Cowan. "Evolution of the Wilson–Cowan equations". In: *Biological cybernetics* 115.6 (2021), pp. 643–653.